

Spot Market Sell Order — Complete Test Steps

File: `tests/trade/spotMarketSellOrder.spec.ts` **Data:** `src/data/spotMarketSellData.csv` **Mode:** Serial (each step depends on the previous one)

SETUP — Before All Tests

Step 1: Launch a Chromium browser (non-headless, 1280×720 viewport) and open a new browser page.

Step 2: Navigate to the login page (`BASE_URL` + login route).

Step 3: Enter the email address in the email field and click "Continue".

Step 4: Enter the password in the password field and click "Continue".

Step 5: Read the current TOTP/OTP value and enter it in the OTP field, then click "Continue" to complete login successfully.

Step 6: Dismiss any post-login dialogs (popups, banners, announcements) and wait for the home/dashboard page to be fully loaded.

TC-01 — Navigate to Spot Trading

Step 7: Click on the Spot Trading link/menu item to navigate to the Spot Trading page and wait for it to load.

TC-02 — Verify Page Labels

Step 8: Read all visible labels on the Spot Trading page and verify each one is correct:

- "Depth View" label is present
 - "Order Book" heading is present
 - "Buy" tab is present
 - "Sell" tab is present
 - "Limit" tab is present
 - "Market" tab is present
 - "Market Trades" label is present
 - "My Trades" label is present
 - "All Orders" tab is present
 - "Trade History" label is present
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TC-03 — Search Currency Pair

Step 9: Type the pair name (e.g. `BTC/USDT`, read from CSV `searchPair` column) into the search/filter input in the market dropdown and wait for results to appear.

TC-04 — Mark as Favorite

Step 10: Click the star/favorite icon next to the pair in the search results to mark it as a favorite.

Step 11: Verify the success toast/status message confirms the pair was **added** to favorites.

TC-05 — Unmark from Favorites

Step 12: Click the star/favorite icon again to unmark the pair from favorites.

Step 13: Verify the status message confirms the pair was **removed** from favorites.

TC-06 — Select Currency Pair

Step 14: Click the "ALL" tab in the pair list, find the pair (e.g. BTC/USDT), and click it to activate it as the active trading pair. The whole page (order book, ticker, trade form) now shows data for this pair.

TC-02b — Verify 24h Ticker vs Binance

Step 15: Simultaneously fetch the 24h ticker data from the page and from the Binance public API for the same pair, then verify:

- Page last price is a positive number
 - Page last price is within 1% of Binance last price (live price tolerance)
 - Page 24h high matches Binance exactly
 - Page 24h low matches Binance exactly
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TC-03b — Order Book Column Headers

Step 16: Read the order book column headers and verify all three are visible and non-empty: **Price, Amount, Total**.

TC-03c — Order Book View Switches

Step 17: Switch the order book to "Sell-only" view → verify the order book is still visible and the top ask price is positive.

Step 18: Switch to "Buy-only" view → verify the order book is still visible and the top bid price is positive.

Step 19: Switch back to "All" view → verify both top ask and top bid prices are positive.

TC-03d — Order Book Precision Dropdown

Step 20: Cycle through precision values **0.01 → 0.1 → 1 → 0.01** (restoring the default). For each precision, and for each of the three views (all / sell / buy), verify that displayed prices have no more decimal places than the precision allows:

- 0.01 → max 2 decimal places
- 0.1 → max 1 decimal place
- 1 → max 0 decimal places

Restore to "All" view at 0.01 precision at the end.

TC-03e — Order Book LTP & Buy/Sell Ratio (Suggestions Only)

Step 21: Attempt to read the Last Traded Price (LTP) from the order book mid-row. Record as a **suggestion** that LTP display is not yet implemented. No assertion failure.

Step 22: Attempt to read the buy % and sell % ratio bar values. Record as a **suggestion** that the ratio bar is not yet implemented. No assertion failure.

TC-03f — Order Book Has Real Data Rows

Step 23: Count the ask rows and bid rows in the order book, read the top ask price and top bid price. Verify:

- Ask row count > 0
 - Bid row count > 0
 - Top ask price > 0
 - Top bid price > 0
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TC-03g — Valid Spread

Step 24: Read the top bid and top ask prices. Verify that top bid < top ask (a valid market spread exists).

TC-03h — Bid/Ask vs Binance within 0.5%

Step 25: Simultaneously fetch the top bid/ask from the page and from the Binance order book API. Verify:

- Page top bid is within 0.5% of Binance top bid
 - Page top ask is within 0.5% of Binance top ask
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TC-03i — Order Book LTP vs Binance within 0.5%

Step 26: Simultaneously read the page order book LTP and the Binance 24h last price. Verify the LTP is positive and within 0.5% of the Binance last price.

TC-06a — Market Sell Tab Price Field is Disabled

Step 27: Click the **Sell** tab, then click the **Market** tab to activate the Market Sell form.

Step 28: Verify the Price input field is **disabled / read-only** (on Market orders the price is determined by the market, not entered by the user).

TC-06a1 — Sell Button Label

Step 29: Read the text on the Sell button. Verify it contains "SELL" and the base coin name (e.g. "SELL BTC").

TC-06a2 — Sell Available Balance is Non-Negative

Step 30: Read the "Available" balance shown next to the sell form (base coin, e.g. BTC). Verify it is ≥ 0 .

TC-06a3 — 25% Button

Step 31: Click the **25%** button. Verify the amount field is filled with a positive value close to 25% of the available BTC balance (within 5% tolerance).

TC-06a4 — 50% Button

Step 32: Click the **50%** button. Verify the amount field is filled close to 50% of the available BTC balance (within 5% tolerance).

TC-06a5 — 75% Button

Step 33: Click the **75%** button. Verify the amount field is filled close to 75% of the available BTC balance (within 5% tolerance).

TC-06a6 — 100% Button

Step 34: Click the **100%** button. Verify the amount field is filled close to 100% of the available BTC balance (within 5% tolerance).

TC-06a7 — Pair Header Shows Selected Pair

Step 35: Read the pair header/title shown above the order form. Verify it contains the base coin name (e.g. "BTC") or the full pair name (e.g. "BTC/USDT").

TC-06b — Capture Pre-Order Balance Snapshot

Step 36: Navigate to the **Portfolio** → **Spot** page and capture a full balance snapshot including:

- Available BTC balance (sell side)
- Available USDT balance (buy side)
- Portfolio spot balances for both BTC and USDT
- Funds balances for both BTC and USDT

Navigate back to the Spot Trading page. Verify the sell available balance is ≥ 0 and matches the portfolio spot balance.

TC-07 — Enter Sell Amount and Verify Estimated Fee

Step 37: Click the **Market** tab to ensure it is active.

Step 38: Clear the amount field and type the sell amount (e.g. `0.0001` BTC — value from CSV `sellAmount` column).

Step 39: Read the estimated fee shown on the form. Calculate the expected fee as: $\text{sellAmount} \times \text{currentMarketPrice} \times \text{feePercent} \div 100$ (feePercent from CSV). Verify the UI estimated fee matches the calculated fee. If the fee element is not displayed, log a suggestion and skip the assertion.

TC-08 — Note BTC Balance Before Placing Order

Step 40: Read and log the current BTC available balance (from the snapshot captured in TC-06b). Verify it is ≥ 0 . This is the baseline for the post-order comparison.

TC-09 — Place Market Sell Order and Verify Success

Step 41: Click the **Market** tab (re-activate to ensure a clean state), then re-enter the sell amount from CSV.

Step 42: Check if the text "Insufficient balance" is visible on the form within 2 seconds.

- If **YES** → mark order as failed, record a soft failure with the message "Insufficient balance — top up and re-run", and **stop this test**. All remaining tests (TC-09b through TC-14b) will be skipped.

Step 43: If balance is sufficient → record the current timestamp as `orderPlacedAt`.

Step 44: Click the **Sell BTC** button to submit / open the confirmation dialog.

Step 45: In the confirmation dialog/modal, read the executed price, executed amount, and order ID. Click **Confirm** to finalize the order.

Step 46: Wait for the success message/toast. Verify the message contains "success", "created", "placed", or "status:". If it does → `orderSucceeded = true`.

TC-09b — Balance Snapshot After Fill

Skipped if TC-09 failed or balance was insufficient.

Step 47: Wait 2 seconds for the exchange to process the fill.

Step 48: Navigate to Portfolio and capture a new full balance snapshot.

Step 49: Calculate expected balance changes:

- BTC should **decrease** by `sellAmount`
- USDT should **increase** by $\text{sellAmount} \times \text{executedPrice}$

Step 50: Compare the before-snapshot (TC-06b) vs the after-snapshot for all of the following, each within 0.5% tolerance:

- Sell available balance (BTC)
- Buy available balance (USDT)
- Portfolio BTC spot balance
- Portfolio USDT spot balance
- Funds BTC amount
- Funds USDT amount

If a coin is not found in the portfolio, log a suggestion instead of failing.

TC-09c — Market Trades Panel Has Correct Headers

Step 51: Click the **Market Trades** tab in the bottom panel. Read the column headers. Verify they include "price", "amount" (or "qty"), and "time".

TC-09d — My Trades Panel Has Correct Headers

Step 52: Click the **My Trades** tab. Read its column headers. Verify they include "price", "amount" (or "qty"), and "time". Switch back to Market Trades tab.

TC-09e — Market Trades Has Real Data Rows

Step 53: Read the data rows in the Market Trades panel. Verify:

- At least 1 row exists
 - The first row's price is positive
 - The first row's amount is positive
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TC-09f — Market Trades Prices Match Binance within 1%

Step 54: Simultaneously read the Market Trades rows from the page and fetch the 20 most recent trades from the Binance API. Verify the most recent page trade price is within 1% of the Binance most recent trade price.

TC-10 — Market Sell is NOT in Open Orders

Skipped if TC-09 failed.

Step 55: Open the **Open Orders** tab. Look for a row matching the pair. Verify the sell order is **not present as a pending entry** — market orders fill immediately and should never appear in Open Orders.

TC-11 — All Orders Shows Market Sell as Filled

Skipped if TC-09 failed.

Step 56: Open the **All Orders** tab. Find the latest row. Verify:

- Pair column contains "BTCUSDT" (slash-stripped comparison)

- Side contains "sell"
 - Type contains "market"
 - Status matches **done** / **filled** / **complete**
 - Executed price > 0
 - Filled amount > 0
 - Remaining = 0
 - Total > 0
 - Date/time is within 60 seconds of when the order was placed (**orderPlacedAt**)
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TC-12 — My Trades Shows the Executed Sell Entry

Skipped if TC-09 failed.

Step 57: Read the **My Trades** tab entries. Verify:

- At least 1 entry exists
 - The entry price is positive
 - The entry amount is positive
 - The entry timestamp is within 2 minutes of the order placement time
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TC-13 — BTC Balance Decreased After Market Sell

Skipped if TC-09 failed.

Step 58: Read the current BTC available balance from the trade form. Verify it has decreased compared to the pre-order balance — the **sellAmount** BTC was deducted after the order filled.

TC-14 — Trade History Shows the Market Sell Order

Skipped if TC-09 failed.

Step 59: Open the **Trade History** tab. Find the latest entry matching the order.

- If **no entry found** → log a suggestion ("market order may not appear yet — verify manually") and skip assertions.

Step 60: If entry is found, verify:

- Pair contains "BTCUSDT"
 - Side contains "sell"
 - Executed price > 0
 - Amount > 0
 - Total > 0
 - Fee ≥ 0
 - Date/time is within 60 seconds of the order placement time
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TC-14b — Trade History Bottom Widget Shows the Sell Entry

Skipped if TC-09 failed.

Step 61: Read the first row of the Trade History widget at the bottom of the trading page (this is separate from the full Trade History tab above). Verify:

- Pair references "BTC" or "BTCUSDT"
- Side contains "sell"
- Price is positive

TEARDOWN — After All Tests

Step 62: Close the browser.

Quick Reference Summary

Step Range	Area
1 – 6	Browser launch, login, dismiss dialogs
7	Navigate to Spot Trading page
8	Verify all page labels
9	Search for currency pair
10 – 11	Mark pair as favorite
12 – 13	Unmark pair from favorites
14	Select pair from ALL tab
15	Verify 24h ticker vs Binance
16	Verify order book column headers
17 – 19	Order book view switches (all/sell/buy)
20	Order book precision dropdown
21 – 22	Order book LTP & ratio bar (suggestions)
23 – 24	Order book real rows and valid spread
25 – 26	Order book bid/ask/LTP vs Binance
27 – 28	Market Sell tab Price field is disabled
29	Sell button label is correct
30	Sell available balance is non-negative
31 – 34	25% / 50% / 75% / 100% percentage buttons
35	Pair header shows selected pair
36	Capture pre-order full balance snapshot

Step Range	Area
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40	Note BTC balance before placing order
41 – 46	Place market sell order and verify success
47 – 50	Post-fill balance snapshot comparison
51	Market Trades panel headers
52	My Trades panel headers
53	Market Trades has real data rows
54	Market Trades prices vs Binance within 1%
55	Market sell NOT in Open Orders
56	All Orders shows sell as Filled
57	My Trades shows executed sell entry
58	BTC balance decreased after sell
59 – 60	Trade History full tab validation
61	Trade History bottom widget validation
62	Close browser